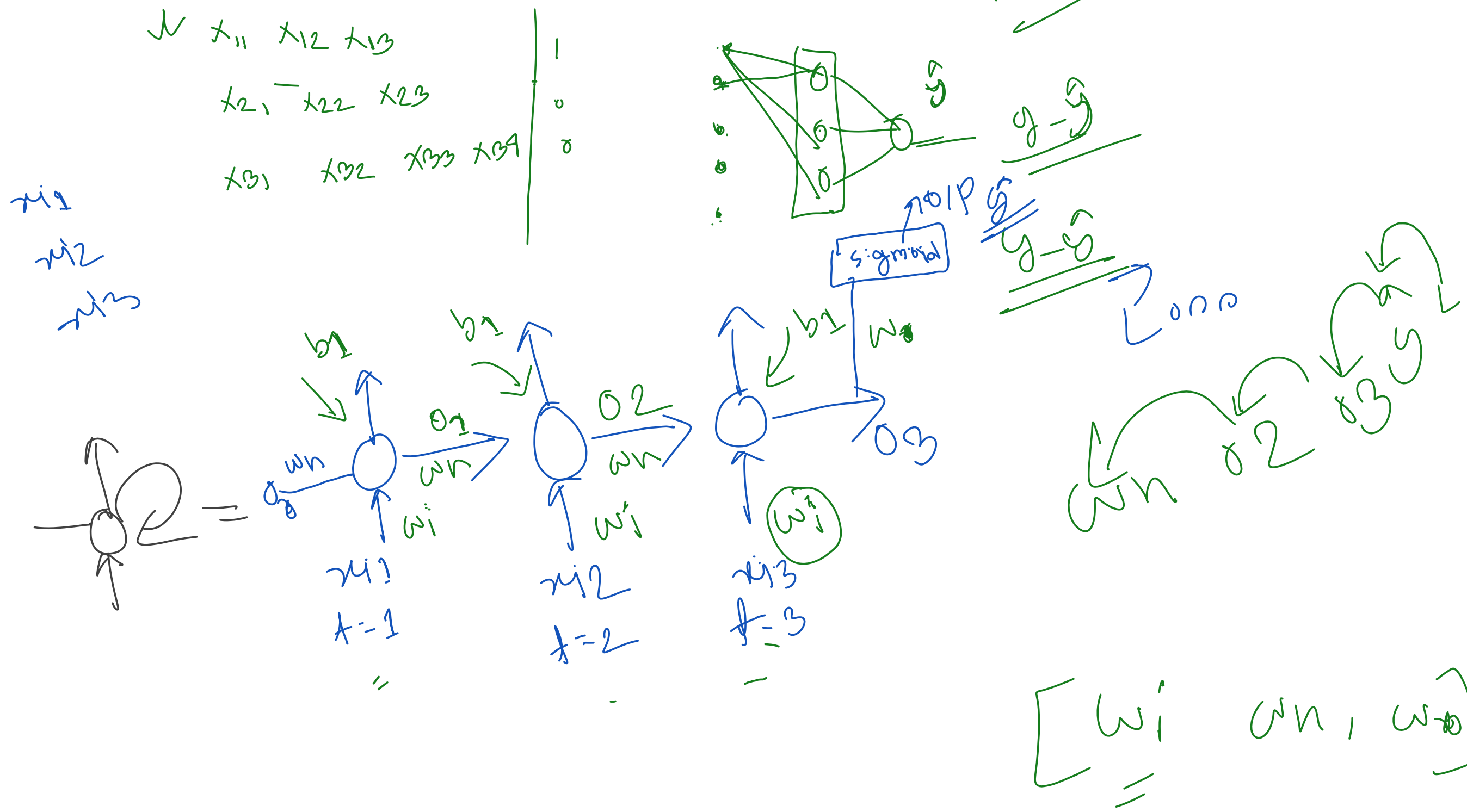


Backprop: Simple Path

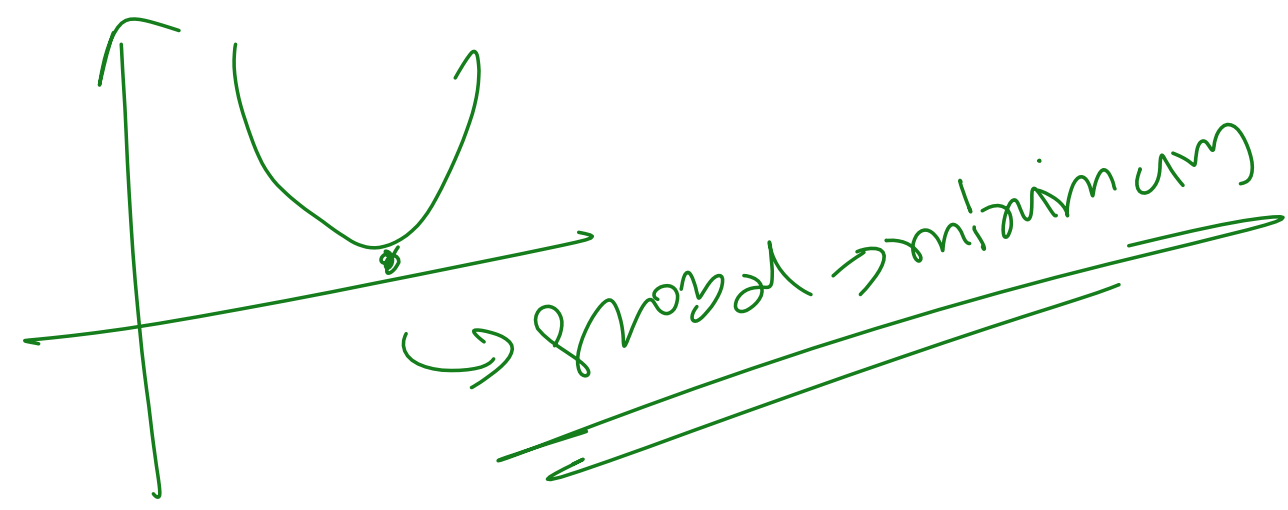


forward propagation

$$\begin{aligned} o_1 &= f(x_{11} \cdot w_i + o_0 \cdot w_n + b_1) \\ o_2 &= f(x_{12} \cdot w_i + o_1 \cdot w_n + b_2) \\ o_3 &= f(x_{13} \cdot w_i + o_2 \cdot w_n + b_3) \\ \hat{y} &= \sigma(o_3 \cdot w_o) \end{aligned}$$

$\downarrow$
 sigmoid

BPTT



2nd update step

$$[w_i^* \quad w_n \quad w_o]$$

weight update formula

$$w_{new} = w_{old} - \eta \left(\frac{\partial L}{\partial w} \right)$$

η → eta → 0.001

$$\frac{\sigma_L}{\sigma_{w_0}}$$

$$w_{\text{new}} = w_{\text{old}} - \eta$$

$$\frac{\sigma_L}{\sigma_{w_{\text{old}}}}$$

$$\# \frac{\sigma_L}{\sigma_{w_{\text{old}}}}$$

$$w_0 \rightarrow \hat{y} \rightarrow \text{Loss}$$

① update

$$\frac{\sigma_L}{\sigma_{w_{\text{old}}}} = \frac{\sigma_L}{\hat{y}} \times \frac{\sigma_L}{\sigma_{w_{\text{old}}}}$$

② update w_n

for $t=1, t=2, t=3$

$$w_n \rightarrow \hat{y} \rightarrow \text{Loss}$$

w_n

$\frac{\sigma_L}{\sigma_{w_{\text{old}}}}$

$$w_n \rightarrow \hat{y} \rightarrow \text{Loss}$$

$$w_{\text{new}} = w_{\text{old}} - \eta \left[\frac{\partial L}{\partial w_{\text{old}}} \right]$$

$$\# \frac{\partial L}{\partial w_{\text{old}}} = \left[\frac{\partial L}{\partial w_{\text{old}}} \right]_{t=2}$$

$$w_n \rightarrow 0.1 \rightarrow 0.2 \rightarrow \dots$$

$$\Rightarrow \left[\frac{\partial L}{\partial y} \times \frac{\partial y}{\partial x} \times \frac{\partial x}{\partial w_n} \right] + \left[\frac{\partial L}{\partial y} \times \frac{\partial y}{\partial x} \times \frac{\partial x}{\partial w_n} \right] + \left[\frac{\partial L}{\partial y} \times \frac{\partial y}{\partial x} \times \frac{\partial x}{\partial w_n} \right] + \left[\frac{\partial L}{\partial y} \times \frac{\partial y}{\partial x} \times \frac{\partial x}{\partial w_n} \right]$$

Long term

Short term

$t=1$ $t=2$ $t=3$ $\dots$

$$\textcircled{2} w_i \text{ time steps: } t=1, 2, 3$$

$$w_{\text{new}} = w_{\text{old}} - \eta \left[\frac{\partial L}{\partial w_{\text{old}}} \right]$$

$$\left[w_{\text{new}} = w_{\text{old}} - \eta \left[\frac{\partial L}{\partial w_{\text{old}}} \right] \right]_{t=3}$$

$$w_i \rightarrow 0.1 \rightarrow 0.2 \rightarrow 0.3 \rightarrow 0.4 \rightarrow 0.5$$

$$t=4 \rightarrow w_i \rightarrow 0.1 \rightarrow 0.2 \rightarrow 0.3 \rightarrow 0.4$$

$$\Rightarrow \frac{\partial L}{\partial w_{jold}} = \left[\frac{\partial L}{\partial y} \times \frac{\partial y}{\partial z} \times \frac{\partial z}{\partial w_{jold}} \right] + \left[\frac{\partial L}{\partial y} \times \frac{\partial y}{\partial z} \times \frac{\partial z}{\partial w_{jold}} \right] + \left[\frac{\partial L}{\partial y} \times \frac{\partial y}{\partial z} \times \frac{\partial z}{\partial w_{jold}} \right]$$

